Clear Form

BOISE STATE UNIVERSITY
OFFICE OF THE REGISTRAR

Application for Graduate Independent Study

Thomas Lonowski	04/20/2026
Student Name	Date
114184926	thomaslonowski@u.boisestate.edu
Student ID	BroncoMail Email

M.S. Mathematics	2025-2026	FA27
Graduate Program	Catalog year	Anticipated Graduation Date

Instructions

Purpose: A Graduate Independent Study involves advanced study of a specialized topic. Any student who has been admitted to the Graduate College is eligible. The graduate student is expected to work under the supervision of a member of the graduate faculty but with a high degree of autonomy.
Note: Independent studies cannot substitute for regularly scheduled course offerings. Nor may a scheduled course offering be a significant component of an independent study.

Procedure: The student is responsible for completing and signing the Application for Graduate Independent Study (GIS) form. Once signed by the Supervising Professor, Advisor, Department Chair, and Program Coordinator the GIS form is to be submitted to the Office of the Registrar for final processing by the posted deadline on the Academic Calendar found here.

Independent Study Course Information

Course and grading information

Semester	Year	Course Prefix	Course #	Credits Requested	Grade	
FA	2026	CS	596	5	☒ Letter grade	☐ Pass/Fail

Infinite Automata, Logic, and Applications
Title of Proposed Independent Study (required)

See attached.
Description of Topic to be Studied by Student (attach a separate sheet, if necessary)

Weekly meetings and a culminating project on advanced topics.
Assessment Method to be Employed by Supervising Professor

Signatures

Thomas Lonowski	Thomas Lonowski	04/20/2026
Student Signature	Print Name	Date

University Use Only

Approved	Disapproved			
☐	☐	Sherman	Elena Sherman	4/21/2026
		Supervising Professor Signature	Print Name	Date
☒	☐	Marion Scheepers	Marion Scheepers	04/21/2026
		Advisor Signature	Print Name	Date
☒	☐	Marion Scheepers	Marion Scheepers	04/21/2026
		Program Coordinator Signature	Print Name	Date
☐	☐	Margaret T. Kinzel	Margaret T. Kinzel	4/22/2026
		Department Chair Signature	Print Name	Date

Deliver to: Boise State Registrar's Office, Administration Building, Room 110, 1910 University Drive, Boise, ID 83725-1365 Email: regmail@boisestate.edu | Phone: (208) 426-4249 | FAX: (208) 426-3169

Course Description: The independent study will be divided into two parts. The first part will cover ω-languages, Monadic Second-Order Logic, Büchi automata, Linear Temporal Logic, determinism and nondeterminism of infinite structures, as well as real-world system modeling and verification, if time allows. The second part will consist of a review of literature and ongoing research problems. If time allows, a workshop publication will be prepared and submitted to an appropriate conference.

Course Text: Automata Theory and Logic by Martin Hofmann (Springer).

Additional Course Resources:

- Principles of Model Checking by Christel Baier and Joost-Pieter Katoen (MIT Press).
- Automata Theory and its Applications by Bakhadyr Khoussainov and Anil Nerode.
    - Focuses more on infinite graph applications rather than system verification.
- On the Translation of Automata to Linear Temporal Logic (ETAPS conference literature).
    - Open access, useful for converting between automata and LTL.

Conferences:

- CIAA: International Conference on Implementation and Application of Automata
    - More theory-focused
    - Palermo, Italy
- Satellite event of ETAPS: International Joint Conferences On Theory and Practice of Software
    - More applied
    - Italy